

Sample Project for Creating Unit Tests

Visual Studio 2013

The "Woodgrove Bank" sample consists of code that you can build into a simple program. You can then generate unit tests that test the various methods, both public and private, of the Woodgrove Bank program.

This sample code is provided for use in the following walkthroughs:

- [Walkthrough: Creating and Running Unit Tests for Managed Code](#) . This walkthrough leads you through the steps to create and customize unit tests, run them, and examine the test results.
- [Walkthrough: Run tests and view code coverage](#) . This walkthrough illustrates how to view code coverage data, which shows the proportion of your project's code that is being tested.
- [Walkthrough: Using the Command-line Test Utility](#) . In this walkthrough, you use the MSTest.exe command-line utility to run tests and view results.

Note The only intentional error in this sample is that the in Debit method "m_balance += amount" should have a minus not a plus sign before the equals sign.

Sample Code

The most up-to-date code for this sample is available here:

```
using System;

namespace BankAccountNS
{
    /// <summary>
    /// Bank Account demo class.
    /// </summary>
    public class BankAccount
    {
        private string m_customerName;

        private double m_balance;

        private bool m_frozen = false;

        private BankAccount()
        {
        }

        public BankAccount(string customerName, double balance)
        {
            m_customerName = customerName;
            m_balance = balance;
        }

        public string CustomerName
        {
            get { return m_customerName; }
        }

        public double Balance
        {
            get { return m_balance; }
        }

        public void Debit(double amount)
        {
            if (m_frozen)
            {
                throw new Exception("Account frozen");
            }

            if (amount > m_balance)
            {
                throw new ArgumentOutOfRangeException("amount");
            }

            if (amount < 0)
            {
                throw new ArgumentOutOfRangeException("amount");
            }

            m_balance += amount;
        }

        public void Credit(double amount)
        {
            if (m_frozen)
```

```

        {
            throw new Exception("Account frozen");
        }

        if (amount < 0)
        {
            throw new ArgumentOutOfRangeException("amount");
        }

        m_balance += amount;
    }

    private void FreezeAccount()
    {
        m_frozen = true;
    }

    private void UnfreezeAccount()
    {
        m_frozen = false;
    }

    public static void Main()
    {
        BankAccount ba = new BankAccount("Mr. Bryan Walton", 11.99);

        ba.Credit(5.77);
        ba.Debit(11.22);
        Console.WriteLine("Current balance is ${0}", ba.Balance);
    }
}

```

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Working with the Code

To work with this code, you first have to create a project for it in Visual Studio. Follow the steps in the "Prepare the Walkthrough" section in [Walkthrough: Creating and Running Unit Tests for Managed Code](#).

See Also

Tasks

[Walkthrough: Creating and Running Unit Tests for Managed Code](#)

[Walkthrough: Using the Command-line Test Utility](#)

Other Resources

[Walkthrough: Run tests and view code coverage](#)